

Frontend User Journey & Interface Design

User Role-Based Interface Walkthrough

CLIENT User Journey

Page 1: Client Dashboard

URL: `/client/dashboard` User sees:

- Welcome message with company name
- Quick stats: "You have 3 active RFQs, 2 awaiting quotes"
- Recent RFQ status updates

Data displayed:

javascript

```
// From schema: Client → RFQ relationship
{
  activeRfqs: 3,
  pendingQuotes: 2,
  recentRfqs: [
    { id: "rfq_123", projectName: "Steel Building A", status: "QUOTED", createdAt: "2023-10-27",
    { id: "rfq_124", projectName: "Warehouse B", status: "ESTIMATION_IN_PROGRESS", createdAt: "2023-10-27"
  ]
}
```

Actions available:

- "Create New RFQ" button (if hasRfqAccess = true)
- View existing RFQs
- Upload documents

Page 2: Create RFQ Form (if hasRfqAccess = true)

URL: `/client/rfq/create` Form fields:

javascript

```
// Maps to RFQ table
{
  projectName: "Office Complex Phase 1", // Required
  description: "50-story office building with parking", // Text area
  documents: [file1.pdf, file2.dwg], // File upload
  priority: "High/Medium/Low", // Dropdown
  expectedStartDate: "2025-07-01" // Date picker
}
```

After submission:

- RFQ created with status "RECEIVED"
- RFQDocuments created for uploaded files
- Redirect to RFQ detail page

Page 3: RFQ Detail View

URL: `/client/rfq/:id` User sees:

- RFQ basic info (project name, description, date)
- Current status with timeline
- Uploaded documents list
- Estimation details (if available)
- Quote information (if status = QUOTED)

Status timeline display:

- ✓ Received (June 15, 2025)
- ✓ Assigned for Estimation (June 16, 2025)
- 🔄 Estimation in Progress (Current)
- 🕒 Estimation Completed
- 🕒 Quoted
- 🕒 Decision

SALES PERSON User Journey

Page 1: Sales Dashboard

URL: `/sales/dashboard` User sees:

- Active clients without RFQ access
- Recent RFQs created by sales team
- Conversion metrics (RFQ → Awards)

Data displayed:

javascript

```
// From schema: User → RFQ (SalesPersonRFQ relation)
{
  clientsWithoutAccess: [
    { name: "ABC Construction", company: "ABC Corp", hasRfqAccess: false }
  ],
  myRfqs: [
    { projectName: "Industrial Plant", status: "QUOTED", clientName: "XYZ Inc" }
  ],
  thisMonthStats: {
    rfqsCreated: 12,
    quoted: 8,
    awarded: 3
  }
}
```

Page 2: Create RFQ for Client

URL: Form sections:

Client Selection:

javascript

```
// If client exists
clientId: "select from dropdown",
// If new client
clientDetails: {
  name: "John Smith",
  email: "john@company.com",
  company: "Smith Construction",
  phone: "+1-555-0123"
}
```

Project Details:

javascript

```
// Same as client RFQ creation
projectName: "Retail Complex",
description: "Multi-story retail with basement parking",
documents: [files...],
salesNotes: "High priority client, existing relationship"
```

After submission:

- Client created (if new) with hasRfqAccess = false
 - RFQ created with salesPersonId = current user
 - Email notification to department manager
-

DEPARTMENT MANAGER User Journey

Page 1: Department Manager Dashboard

URL: /manager/dashboard User sees:

- RFQs pending assignment
- Active estimations overview
- Team workload distribution
- Performance metrics

Data displayed:

javascript

// Complex queries across multiple tables

```
{
  pendingAssignment: [
    {
      rfqId: "rfq_123",
      projectName: "Office Building",
      clientName: "ABC Corp",
      receivedDate: "2025-06-15",
      priority: "High",
      salesPerson: "John Sales"
    }
  ],
  activeEstimations: [
    {
      estimationId: "est_456",
      projectName: "Warehouse Project",
      assignedTo: "Estimation Head A",
      status: "IN_PROGRESS",
      dueDate: "2025-06-25"
    }
  ],
  teamWorkload: [
    { estimatorName: "John Est", activeTasks: 3, avgHoursPerWeek: 35 },
    { estimatorName: "Jane Head", activeTasks: 5, avgHoursPerWeek: 42 }
  ]
}
```

Page 2: RFQ Assignment Form

URL: `/manager/rfq/:id/assign` **Form displays:**

- RFQ details and documents
- Available estimators/estimation heads dropdown
- Priority setting
- Due date
- Special instructions

Assignment options:

javascript

// Two assignment paths from schema

```
assignTo: {  
  type: "ESTIMATION_HEAD", // or "DEPT_MANAGER" (self)  
  userId: "user_789",  
  priority: "High",  
  dueDate: "2025-06-30",  
  instructions: "Focus on connection details"  
}
```

After assignment:

- Estimation record created
- EstimationTask created with assignedToId
- RFQ status updated to "ASSIGNED_FOR_ESTIMATION"

Page 3: Estimation Review Page

URL: `/manager/estimation/:id/review` **User sees:**

- Complete estimation details
- Line items with calculations
- Individual estimator contributions
- Time spent analysis

Review actions:

javascript

```
// Three options as per workflow
reviewAction: {
  type: "APPROVE", // or "REJECT" or "EDIT"
  comments: "Looks good, ready for CEO pricing",
  // If EDIT
  adjustments: {
    totalHours: 120, // Modified from original 150
    totalWeeks: 3,  // Modified from original 4
    reason: "Based on similar recent project"
  }
}
```

ESTIMATION HEAD User Journey

Page 1: Estimation Head Dashboard

URL: /estimation-head/dashboard User sees:

- Assigned estimations (can work on personally)
- Team members available for assignment
- Pending reviews from team
- Personal + team performance metrics

Data displayed:

javascript

```
{
  myAssignments: [
    {
      estimationId: "est_123",
      projectName: "Steel Bridge",
      assignedBy: "Dept Manager",
      dueDate: "2025-06-28",
      canDelegate: true
    }
  ],
  teamMembers: [
    {
      name: "Junior Estimator 1",
      currentTasks: 2,
      availability: "Available",
      efficiency: "85%"
    }
  ],
  pendingReviews: [
    {
      taskId: "task_456",
      estimatorName: "Junior Est 2",
      projectName: "Warehouse",
      completedDate: "2025-06-18"
    }
  ]
}
```

Page 2: Estimation Assignment Decision

URL: `/estimation-head/estimation/:id` Two options presented:

Option 1: Work Personally

- "Start Estimation" button
- Direct redirect to estimation form

Option 2: Delegate to Team

- Team member selection dropdown
- Assignment form with instructions
- Due date setting

Page 3: Estimation Form (if working personally)

URL: `/estimation-head/estimation/:id/work` Form sections:

Basic Details:

```
javascript

// Maps to EstimationTemplate
templateData: {
  fabricatorName: "Steel Works Inc",
  teklaOrSds: "TEKLA", // Dropdown
  connectionDesignRequired: true, // Checkbox
  scopeSheetAvailable: false, // Checkbox
  steelConsidered: "Main & Misc. Steel"
}
```

Line Items Section:

```
javascript

// Dynamic form for EstimationLineItem
lineItems: [
  {
    itemDescription: "Main Structural Steel",
    quantity: 150,
    unit: "Tons",
    hoursPerUnit: 8.5, // From Excel formulas
    totalHours: 1275, // Calculated
    weeks: 2.3 // Calculated
  },
  {
    itemDescription: "Connection Design",
    quantity: 200,
    unit: "Connections",
    hoursPerUnit: 2.1,
    totalHours: 420,
    weeks: 0.75
  }
]
```

Excel Integration Panel:

- "Import from Excel" button
- Formula calculator for complex items
- Historical data reference

ESTIMATOR User Journey

Page 1: Estimator Dashboard

URL: `/estimator/dashboard` User sees:

- Assigned tasks
- Personal efficiency metrics
- Recent completed work
- Knowledge base access

Data displayed:

javascript

```
// From EstimationTask where assignedToId = current user
{
  activeTasks: [
    {
      taskId: "task_789",
      estimationId: "est_123",
      projectName: "Retail Complex",
      assignedBy: "Estimation Head A",
      assignedDate: "2025-06-16",
      dueDate: "2025-06-23",
      status: "IN_PROGRESS",
      hoursSpent: 15.5
    }
  ],
  personalMetrics: {
    thisMonth: {
      tasksCompleted: 8,
      avgHoursPerTask: 18.2,
      onTimeDelivery: "87%",
      accuracyRate: "92%"
    }
  }
}
```

Page 2: Task Work Interface

URL: `/estimator/task/:id` User sees:

- Project documents viewer
- Estimation form (similar to Estimation Head)
- Time tracking widget
- Progress saving capability

Time Tracking:

javascript

```
// Updates EstimationTask.hoursSpent
timeTracking: {
  startTime: "2025-06-19 09:00",
  breaks: [
    { start: "12:00", end: "13:00" } // Lunch
  ],
  currentSession: 4.5, // hours
  totalTaskTime: 20.0 // hours
}
```

Form Auto-save:

- Saves to EstimationLineItem every 5 minutes
- Status remains "IN_PROGRESS" until submission

Page 3: Task Submission

URL: `/estimator/task/:id/submit` Final review page:

- Line items summary
- Total hours/weeks calculation
- Upload supporting documents
- Add notes for reviewer

Submission data:

javascript

```
submissionData: {
  finalHours: 125.5,
  finalWeeks: 3.2,
  notes: "Connection design may need PE review",
  confidence: "High", // Self-assessment
  supportingDocs: ["calc_sheet.pdf"]
}
```

After submission:

- Task status → "COMPLETED"
- Estimation status → "PENDING_REVIEW"
- Notification to Estimation Head/Dept Manager

ADMIN (CEO) User Journey

Page 1: Executive Dashboard

URL: `/admin/dashboard` **User sees:**

- High-level business metrics
- Estimations pending pricing
- Revenue pipeline
- Team performance overview

Data displayed:

javascript

```
{
  businessMetrics: {
    totalRfqsThisMonth: 25,
    quotedProjects: 18,
    awardedProjects: 7,
    pipelineValue: "$2,450,000",
    winRate: "38.9%"
  },
  pendingPricing: [
    {
      estimationId: "est_456",
      projectName: "Industrial Complex",
      clientName: "Major Corp",
      estimatedHours: 340,
      deptManagerApproval: "2025-06-18",
      urgency: "High"
    }
  ],
  teamPerformance: {
    topPerformers: [
      { name: "Senior Estimator A", efficiency: "94%", accuracy: "96%" }
    ],
    avgEstimationTime: "3.2 days",
    avgAccuracy: "89%"
  }
}
```

Page 2: Pricing Interface

URL: `/admin/estimation/:id/price` **User sees:**

- Complete estimation breakdown
- Historical pricing data for similar projects
- Market rate calculator
- Profit margin settings

Pricing form:

javascript

```
pricingData: {  
  // From EstimationTemplate - read only  
  detailingEstimatedHours: 340,  
  detailingEstimatedWeeks: 8.5,  
  
  // CEO inputs  
  hourlyRate: 125, // $/hour  
  detailingPrice: 42500, // Calculated or manual  
  
  mainSteelConnectionPrice: 15000,  
  miscSteelDesignPrice: 8500,  
  
  profitMargin: "22%",  
  finalQuoteAmount: 66000,  
  
  validityPeriod: 30, // days  
  paymentTerms: "Net 30",  
  specialTerms: "PE stamping included"  
}
```

Page 3: Quote Generation & Approval

URL: Final quote preview:

- Professional formatted quote using EstimationTemplate data
- All inclusions/exclusions listed
- Terms and conditions
- Digital signature area

Actions:

- "Send Quote" - updates RFQ status to "QUOTED"
- "Save Draft" - keeps for later review
- "Request Changes" - sends back to Dept Manager

Cross-Role Interface Elements

Notification System

All roles receive contextual notifications:

Department Manager:

- "New RFQ #123 received from ABC Corp"
- "Estimation #456 completed by John Estimator"

Estimation Head:

- "New assignment: Steel Bridge project due June 30"
- "Team member completed Warehouse estimation"

Estimator:

- "New task assigned: Retail Complex connection design"
- "Task #789 approved by manager"

Document Management

Consistent across all roles:

- Drag & drop file upload
- Document viewer with zoom/pan
- Version control for uploaded files
- Category tagging (Drawings, Specs, Calcs)

Reporting Dashboard (Role-based access)

URL:

Admin sees: Business intelligence, P&L impact, win rates **Dept Manager sees:** Team productivity, project timelines **Estimation Head sees:** Team member performance, accuracy rates **Estimator sees:** Personal metrics, skill development

Real-time Updates

Using WebSocket connections:

- Status changes propagate immediately
- Task assignments appear instantly
- Document uploads notify relevant users
- Time tracking updates in real-time

Mobile Responsiveness

All interfaces adapt for mobile/tablet use:

- Simplified navigation for estimators in field
- Document viewer optimized for tablets
- Time tracking accessible on mobile
- Quick status updates for managers

This frontend design ensures each user role has a tailored experience while maintaining data consistency through the underlying schema relationships.